

Harsh Singh

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Professional Summary

Motivated Data Science and AI undergraduate with hands-on experience building end-to-end machine learning pipelines, Retrieval-Augmented Generation (RAG) systems, and data-driven full-stack applications. Proficient in Python, statistical modeling, NLP, and vector search. Experienced in deploying intelligent systems — from LLM-integrated chat applications to IoT edge-computing solutions — that solve real-world problems. Seeking a Data Scientist or ML Engineer role to apply AI/ML expertise in a high-impact engineering environment.

Technical Skills

Languages: Python, C, C++, JavaScript, SQL, HTML, CSS

Machine Learning & AI: Scikit-learn, FAISS, Sentence Transformers, RAG, NLP, LLMs, Semantic Search, Vector Embeddings, Prompt Engineering, OpenAI API

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Chart.js, Statistical Modeling, EDA

Frameworks & Libraries: React.js, Flask, Node.js, Express.js, Tailwind CSS

Embedded & IoT: OpenCV, ESP32, Edge Computing, REST API Integration

Databases & Tools: MySQL, MongoDB, Git, GitHub, VS Code

Core Concepts: Data Structures & Algorithms, DBMS, OOP, Computer Networks, REST APIs, JWT Authentication

Projects

AskPDF — AI-Powered PDF Chat Application | *React.js, FAISS, Sentence Transformers* 2025

- Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline integrating a FAISS vector store and Sentence Transformers, enabling efficient semantic search and question-answering over large, multi-document PDF corpora.
- Engineered advanced document chunking strategies and optimized vector embedding generation, significantly improving retrieval precision and reducing system latency for large-scale document sets.
- Developed a high-performance full-stack application with a React.js ChatGPT-style UI featuring real-time streaming responses, demonstrating applied NLP and production LLM integration.
- Implemented granular source tracking and citation functionality, enabling verifiable AI-generated responses with direct navigation to relevant PDF pages — critical for trustworthy, auditable data-driven outputs.

WealthWave — Smart Personal Finance Manager | *React.js, Node.js, Express.js, MongoDB* 2025

- Designed and developed a full-stack financial analytics platform enabling real-time tracking of income, expenses, budgets, and recurring transactions through an interactive data dashboard.
- Built comprehensive data visualization and reporting features — category-wise charts, monthly trend analysis, and CSV/PDF export — demonstrating applied data analysis and business intelligence engineering.
- Integrated an AI-powered finance assistant capable of answering domain-specific queries, leveraging OpenAI API alongside a local insight inference mode for offline-capable intelligent responses.
- Engineered secure JWT-based authentication with bcrypt password hashing, protected API routes, and custom validation rules, ensuring enterprise-grade data security across the application.

IoT Smart Doorbell & Cloud Monitoring System | *Python, C++, OpenCV, ESP32* 2026

- Developed a real-time edge-computing security system utilizing OpenCV-based facial recognition for sub-second biometric authentication at the point of entry.
- Implemented an asynchronous HTTP web server on ESP32 in C++, handling physical hardware interrupts and local API requests with low-latency response under concurrent load.
- Integrated third-party REST APIs (ImgBB, Blynk IoT) to enable cloud-based image storage, real-time mobile push notifications, and remote access control from a mobile dashboard.
- Designed a fault-tolerant network layer with structured timeout handling and exception recovery, ensuring uninterrupted video stream processing and reliable IoT communication under degraded network conditions.

Certifications & Learning

- **Self-Directed ML/AI Learning:** RAG systems, LLM integration, FAISS vector search, Sentence Transformers, Prompt Engineering, OpenAI API.
- **Data Science Core:** Statistical modeling, Exploratory Data Analysis (EDA), Pandas/NumPy, data pipeline design.

Achievements & Highlights

- Built and deployed a production-grade RAG pipeline (AskPDF) integrating FAISS semantic search with real-time LLM streaming — a stack aligned with industry-standard AI/ML engineering practices.
- Designed a multi-modal AI analytics assistant (WealthWave) combining structured financial data analysis with LLM-powered natural language querying.
- Consistently maintains a CGPA of **8.21** while independently building complex ML, full-stack, and IoT systems.

Research Interests

Large Language Models (LLMs), Retrieval-Augmented Generation, Applied NLP, Generative AI, Data-Centric AI Systems, Semantic Search, Statistical Machine Learning.

Education

GH Rasoni College of Engineering and Management	Pune, India
<i>Bachelor of Technology in Information Technology CGPA: 8.21 / 10</i>	<i>Aug 2023 – Present</i>
KV No-2 Dhar Road, Udhampur	India
<i>Higher Secondary Certificate (HSC) — Science Stream Score: 76%</i>	<i>2022 – 2023</i>
KV No-2 Dhar Road, Udhampur	India
<i>Secondary School Certificate (SSC) Score: 81%</i>	<i>2020 – 2021</i>

Professional Links

LinkedIn: www.linkedin.com/in/harsh-singh-070590298
GitHub: <https://github.com/harshXranger>